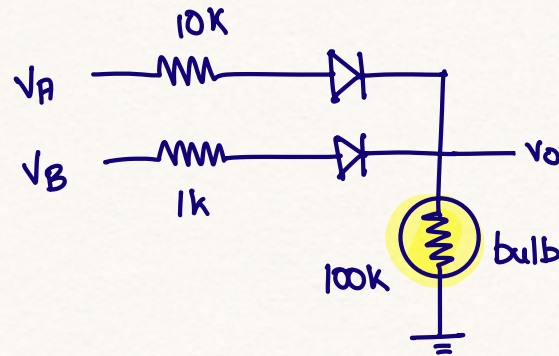


Mid
Solve
Q1 → Set A
Q2 → Set B

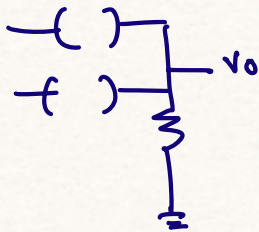
CSE 350
Summer '23



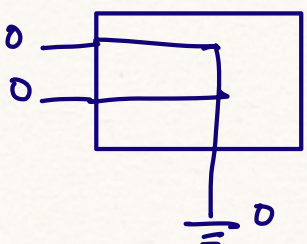
Case I

$$V_A = 0$$

$$V_B = 0$$



$$V_O = 0 \text{ (off)}$$

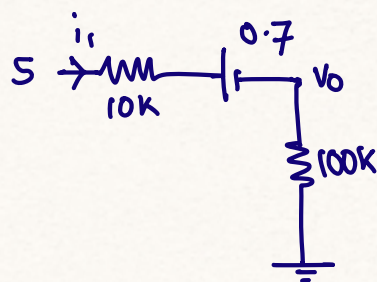


$$P = 0$$

Case II

$$V_A = 5$$

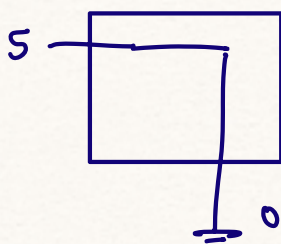
$$V_B = 0$$



$$\frac{V_O}{100k} + \frac{V_O + 0.7 - 5}{10k} = 0$$

$$V_O = 3.91 \text{ V}$$

(dim)



$$i_1 = \frac{3.91}{100k} = 0.0391 \text{ mA}$$

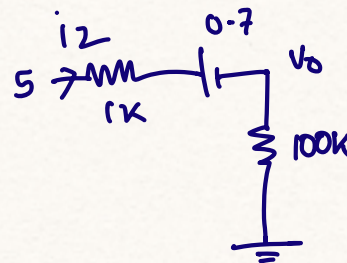
$$P = (5 - 0) 0.0391$$

$$P = 0.1955 \text{ mW}$$

Case III

$$V_A = 0$$

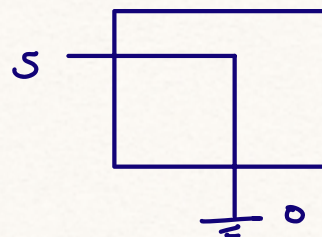
$$V_B = 5$$



$$\frac{V_O}{100} + \frac{V_O + 0.7 - 5}{1k} = 0$$

$$V_O = 4.25 \text{ V}$$

(bright)



$$i_2 = \frac{4.25}{100k} = 0.0425 \text{ mA}$$

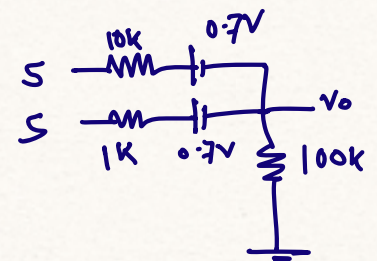
$$P = (5 - 0) 0.0425$$

$$P = 0.2125 \text{ mW}$$

Case IV

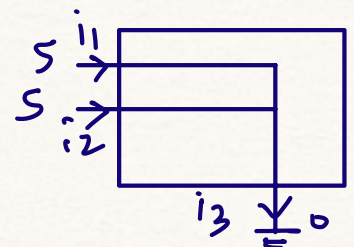
$$V_A = 5$$

$$V_B = 5$$



$$\frac{V_O + 0.7 - 5}{1k} + \frac{V_O + 0.7 - 5}{10k} + \frac{V_O}{100k} = 0$$

$$V_O = 4.26 \text{ V (bright)}$$



$$P = (5 - 0)i_1 + (5 - 0)i_2$$

$$= 5(i_1 + i_2)$$

$$= 5(i_3)$$

$$i_3 = \frac{4.26}{100k} = 0.0426$$

$$P = 0.213 \text{ mW}$$

Set A

state	V_o/V	V_A/V	V_B/V
Bright	4.26 OR 4.25	5 OR 0	5
Dim	3.91	5	0
off	0	0	0
Bright	4.26 OR 4.25	5 OR 0	5

Power

0.213

0.196

0

0.213

0.622 mW

$$\text{Avg Pow} = \frac{0.622}{4} = 0.156 \text{ mW}$$

Set B

state	V_o/V	V_A/V	V_B/V
Dim	3.91	5	0
off	0	0	0
Bright	4.26 OR 4.25	5 OR 0	5
Dim	3.91	5	0

Pow.

0.196

0

0.213

0.196

0.605 mW

$$\text{Avg. Pow} = \frac{0.605}{4} \text{ mW} = 0.151 \text{ mW}$$

Dissipation

- Average Power

⊛ note : allow marks for 4 power calc. in accordance to selection of cases in part(a).

⊛ Bright can be $(S, 0)$ or (S, S) : marks $\rightarrow$ for whichever they select in part (a).

⊛ For set A with two Bright instances they may select $(S, 0)$ AND (S, S) in two cases.

⊛ if they divide by 3 instead of 4 but values of Power for cases are correct deduct 1 mark only.

$$V_{B0} = 0.8, V_o = 0.1,$$

$$V_{c2} = 0.9, V_{B2} = 1.6, V_{B1} = 2.3$$

Set-A-Q4

$$\textcircled{a} \quad I_1 = \frac{6-2.3}{10k} = 0.37 \text{ mA}$$

$$I_{EX} = I_{EY} = \beta_F I_1 = 0.037 \text{ mA}$$

$$I_{B2} = I_1 + 2 I_{EX} = 0.444 \text{ mA}$$

$$I_2 = \frac{6-0.9}{4k} = 1.275 \text{ mA}$$

$$I_{E2} = I_2 + I_{B2} = 1.72 \text{ mA}$$

$$I_{B0} = I_{E2} - I_4 = 1.32 \text{ mA}$$

$$I_3 = \frac{6-0.1}{5k} = 1.18 \text{ mA}$$

Individual Load current,

$$I_L = I'_1 = \frac{6-0.9}{10k} = 0.51 \text{ mA}$$

$$\therefore \frac{I_3 + N I_L}{I_{B0}} \leq \beta_F$$

$$\hookrightarrow \frac{1.18 + N(0.51)}{1.32} \leq 30$$

$$\hookrightarrow N = \lfloor 75.33 \rfloor = \boxed{75}$$

$$\textcircled{b} \quad P = 6(I_1 + I_2 + I_3 + I_{EX} + I_{EY})$$

$$+ 10(0.1)(I_L)$$

$$= \boxed{17.904 \text{ mW}}$$

$$\textcircled{c} \quad \frac{1.18 + 70(0.51)}{1.32} < \beta_F$$

$$\hookrightarrow \boxed{27.939} < \beta_F$$

Set-B-Q3

$$\textcircled{a} \quad I_1 = \frac{5-2.3}{5} = 0.54 \text{ mA}$$

$$I_{EX} = I_{EY} = \beta_F I_1 = 0.054 \text{ mA}$$

$$I_{B2} = I_1 + 2 I_{EX} = 0.648 \text{ mA}$$

$$I_2 = \frac{5-0.9}{2k} = 2.05 \text{ mA}$$

$$I_{E2} = I_2 + I_{B2} = 2.698 \text{ mA}$$

$$I_{B0} = I_{E2} - I_4 = 2.298 \text{ mA}$$

$$I_3 = \frac{5-0.1}{4k} = 1.225 \text{ mA}$$

Individual Load current,

$$I_L = I'_1 = \frac{5-0.9}{5k} = 0.82 \text{ mA}$$

$$\therefore \frac{I_3 + N I_L}{I_{B0}} \leq \beta_F$$

$$\hookrightarrow \frac{1.225 + N(0.82)}{2.298} \leq 40$$

$$\hookrightarrow N = \lfloor 110.603 \rfloor = 110$$

$$\textcircled{b} \quad P = 5 \times (I_1 + I_2 + I_3 + I_{EX} + I_{EY})$$

$$+ 25 \times (0.1)(I_L)$$

$$= \boxed{21.665 \text{ mW}}$$

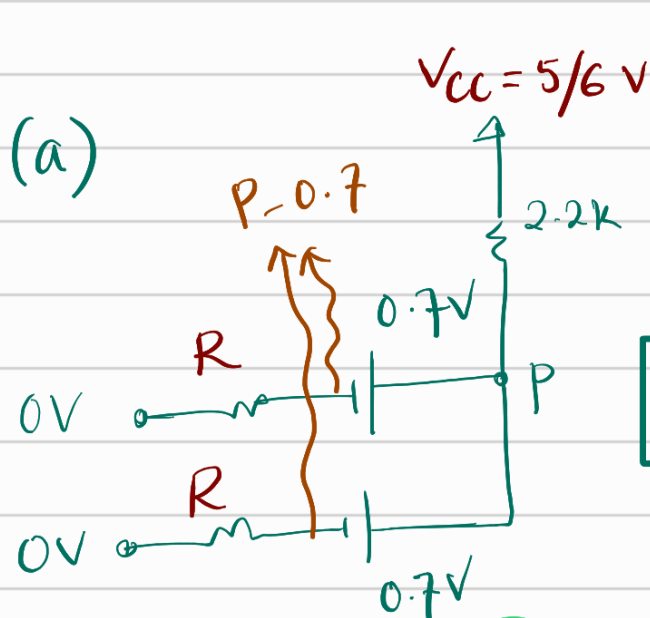
$$\textcircled{c} \quad \frac{1.225 + 110 \times (0.82)}{2.298} < \beta_F$$

$$\hookrightarrow \boxed{39.78} < \beta_F$$

Mid-Summer 2023 (CSE 350)

DTL

$$\{3+1=4\}$$



Nodal Analysis (3)

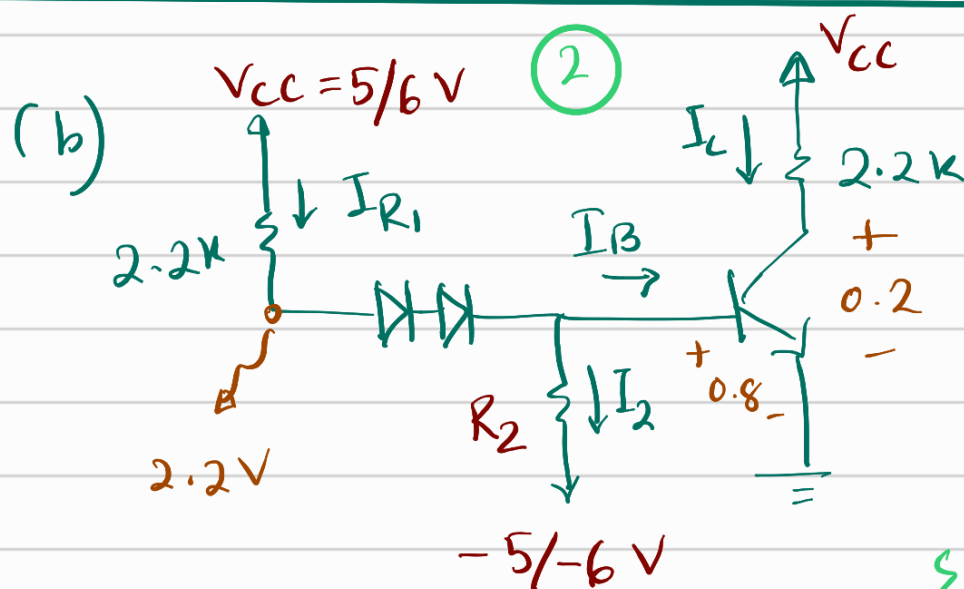
$$\frac{P - V_{cc}}{2.2} + 2 \times \frac{P - 0.7}{R} = 0$$

$$\Rightarrow P = 0.89V \rightarrow \text{Set A}$$

$$\Rightarrow P = 0.98V \rightarrow \text{Set B}$$

$$R = 200/250\Omega$$

(1)



$$[\text{Both A \& B} = 5/6V]$$

$$R_2 = 47/45k\Omega$$

$$\{2+4+1=7\}$$

(4) Set A

$$I_C = \frac{5 - 0.2}{2.2k} = 2.182mA$$

$$I_B = I_{R1} - I_2 = \frac{5 - 2.2}{2.2k} - \frac{0.8 + 5}{47k} = 1.149mA$$

(1)

$$\therefore \beta_{min} \geq I_C / I_B = 1.899 \rightarrow \text{Set A}$$

(4) Set B

$$I_C = \frac{6 - 0.2}{2.2k} = 2.64mA$$

$$I_B = \frac{6 - 2.2}{2.2k} - \frac{0.8 + 6}{45k} = 1.576mA$$

(1)

$$\therefore \beta_{min} \geq I_C / I_B = 1.675 \rightarrow \text{Set B}$$

$$\{4+2+2+1=9\}$$

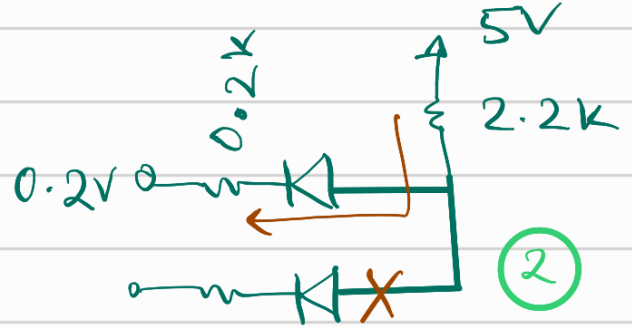
(c) Set A :- $A = B = 5V$ case

$$I_c = I'_c + N I_L; \quad \# \quad I_c = \beta_F \times I_B$$

$$\# \quad I'_c = 2.182 \text{ mA} \quad \left| \quad = 65 \times 1.149 = 74.685 \text{ mA} \right.$$

(4) $\# \quad I_L = \frac{5 - 0.7 - 0.2}{2.2 + 0.2}$

$$= 1.708 \text{ mA}$$



(2) $\therefore N = \lfloor 42.4 \rfloor = 42 \rightarrow \text{set A}$

(Other cases give $N = \infty$) (1)

Set B :- $A = B = 6V$ case

$$\# \quad I_c = 55 I_B = 55 \times 1.576$$

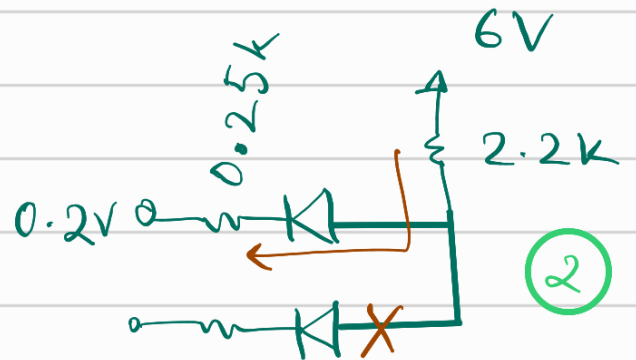
$$= 86.68 \text{ mA}.$$

(4) $\# \quad I'_c = 2.64 \text{ mA}$

$$\{4+2+2+1=9\}$$

$$\# \quad I_L = \frac{6 - 0.7 - 0.2}{2.2 + 0.25}$$

$$= 2.082 \text{ mA}$$



$\therefore N = \lfloor 40.4 \rfloor = 40 \rightarrow \text{set B}$

(2)

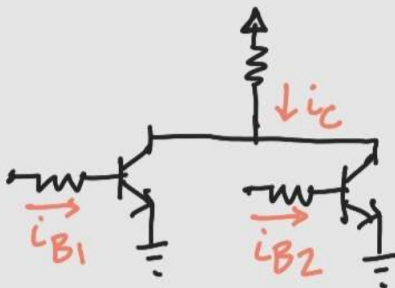
(Other cases give $N = \infty$) (1)

Set-A (Q2)

(a). NOR gate.

V_A	V_B	T_1	T_2	V_O
0	0	cut	cut	5V
0	5	cut	sat.	0.2V
5	0	sat.	cut	0.2V
5	5	sat.	sat.	0.2V

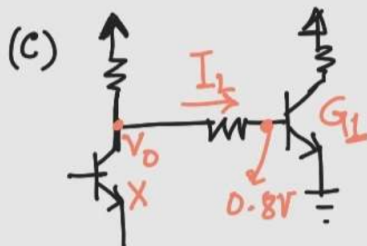
(b)



$$P = i_{B1} \times (5-0) + i_{B2} \times (5-0) + i_c \times (5-0)$$

$$150 = \frac{5-0.8}{0.5} \times 5 + \frac{5-0.8}{0.5} \times 5 + \frac{5-0.2}{R_c} \times 5$$

$$\Rightarrow R_c = \boxed{0.364 \text{ k}\Omega}$$



$$\frac{5-V_0}{R_c} = 7 \times \frac{V_0-0.8}{2}$$

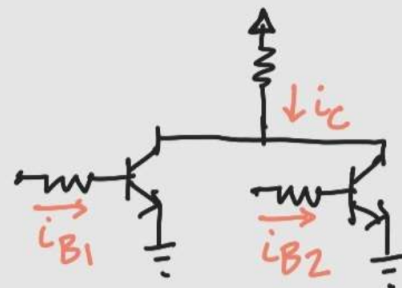
$$\Rightarrow V_0 = \boxed{2.647 \text{ V}}$$

Set-B (Q4)

(a) NOR gate.

V_A	V_B	T_1	T_2	V_O
0	0	cut	cut	5V
0	5	cut		0.2V
5	0		cut	0.2V
5	5	sat.	sat.	0.2V

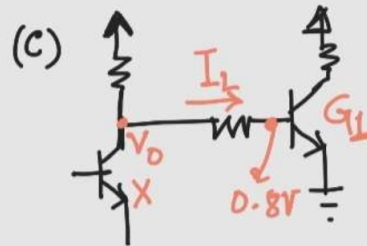
(b)



$$P = i_{B1} \times (5-0) + i_{B2} \times (5-0) + i_c \times (5-0)$$

$$170 = \frac{5-0.8}{0.4} \times 5 + \frac{5-0.8}{0.4} \times 5 + \frac{5-0.2}{R_c} \times 5$$

$$\Rightarrow R_c = \boxed{0.369 \text{ k}\Omega}$$



$$\frac{5-V_0}{R_c} = 6 \times \frac{V_0-0.8}{2.5}$$

$$\Rightarrow V_0 = \boxed{3.027 \text{ V}}$$